COMS 4720 PROJECT

Let us consider an 8 puzzle which has the following goal and can perform single moves such as LEFT, RIGHT, UP, and DOWN, which have a cost of 1, and double moves DBL_LEFT, DBL_RIGHT, DBL_UP, and DBL_DOWN which have a cost of 1, even though it is equivalent to 2 single moves executed back to back :

1	2	3
8	0	4
7	6	5

Since a double move lets the blank travel 2 squares at a time, we can say that the lower for single moves is not admissible by themselves since a tile that is 2 steps away from its goal can be fixed by a single double action and its Manhattan distance would be 2 whereas if we count take single moves then it will take us 2 single moves to reach to the goal and which will equal the Manhattan distance and will not make it a admissible solution.

Let us assume that MD(s) is the standard Manhattan distance for s and MT is the number of misplaced tiles. We know that each single step action can reduce MD by 1 and each double step action can reduce MD by at most in a single action.Thus, we can say that the number of actions needed from any state is atleast [MD/2] and similarly, we can say that a double move can place atmost 2 misplaced tiles correctly, hence we can say that the actions needed from any state is atleast [MT/2].

Since we assume that the both MT and MD are admissible then that means that the true cost must be as large as both heuristics individually, thus if we have to take out an admissible estimate, then we must take out its maximum. Thus, if we consider that h_3 is an admissible heuristic function, then it can be defined as follows:

h_3 = max(MD/2,MT/2)